ing technologies. The following sections cover some of the key technology ingredients for the digital home design guidelines.

The IPv4 family of protocols is the foundation for networking and connectivity in the digital home. IP also provides the underlying network communications for devices on the Internet. IP is based on industry-standard specifications implemented and supported in a wide range of devices with more than two decades of deployment in government, academic, and commercial environments.

There are several advantages to using IP in the digital home:

- IP allows applications running over diff e rent media to commnicate transparently. IP will run over many different media with out any awareness required by applications as to the underlying media. For example, a PC or advanced STB may stream media content to a TV in the master bedroom through an Ethernet cable to an 802.11 Access Point, and then wirelessly to the TV. With IP, the media server and the TV are unaware that the media content travels over two separate physical media. For direct peer-to-peer communications of a mobile device transmitting to a digital home device, IP provides the unifying framework to make applications independent from the actual transport technology.
- IP allows connecting every device in the home to the Internet. Since IP is the protocol of the Internet, any device in the digital home can be potentially connected to any other Internet-connected device in the world.
- IP connectivity is inexpensive. Because it is ubiquitous, economies of scale and competition combine to make IP physical media implementations available at lower cost than other technologies.

Recognising these advantages, the design guidelines for networking and connectivity are intended to facilitate simple, interoperable connectivity, while meeting the digital home's needs today and into the future.

The IETF is standardising IPv6 as an improved version of IP and is actively pursuing a range of transition techniques for a smooth migration from IPv4 to IPv6. Many of these techniques will be applicable to home devices and residential gateways.